

Hands-on Lab: Automating Fabric Data Agent using SDK

Lab Duration: 30 - 45 Minutes

Overview

In this lab, you will learn how to automate the creation and management of Data Agents in Microsoft Fabric using the Fabric Data Agent SDK.

You will programmatically create, configure, and manage a Data Agent, connect it to a Lakehouse, control schema usage, and enhance query generation using instructions and few-shot examples.

Learning Objectives

After completing this lab, you will be able to:

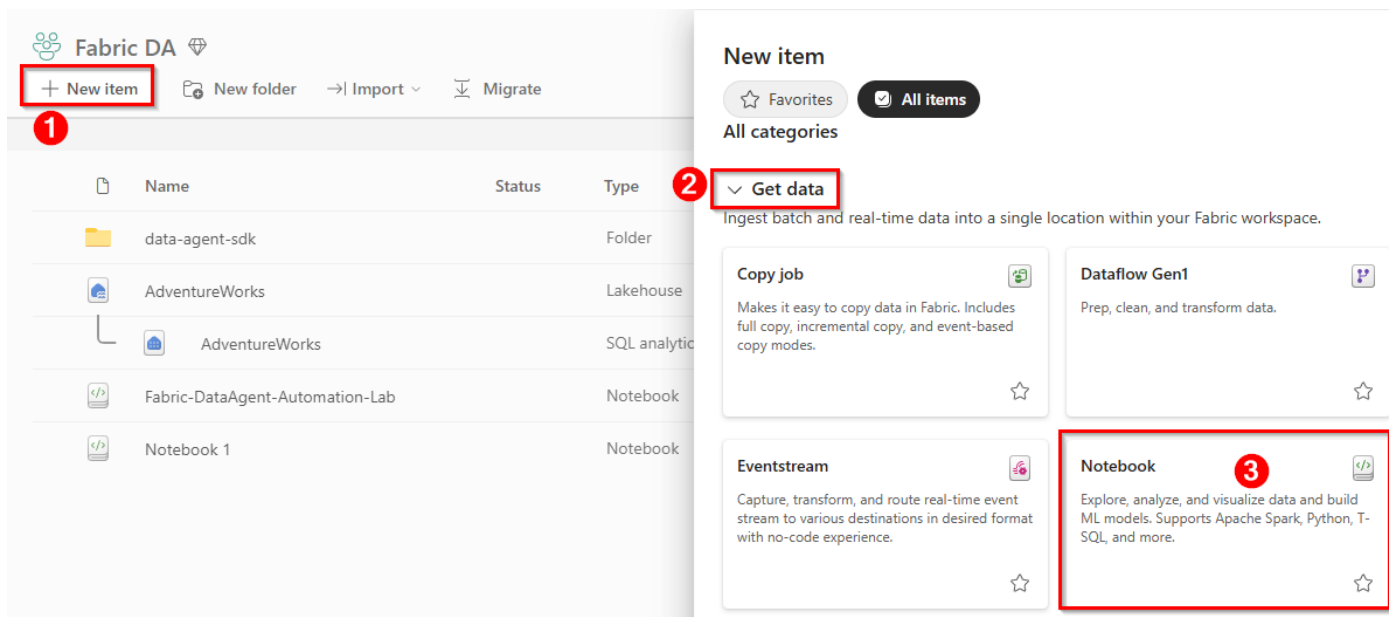
- Create and manage Data Agents programmatically
- Configure AI instructions
- Attach and manage datasources
- Control table-level query behavior
- Add few-shot examples for better query responses
- Publish and delete Data Agents

Prerequisites

- Access to Microsoft Fabric workspace
- A Lakehouse (recommended: **AdventureWorks**)
- Basic knowledge of Python
- Notebook access in Fabric

Task 1: Create Notebook Environment

1. Navigate to Fabric Workspace
2. Click on **New** → **Notebook**



3. Rename the notebook: **Fabric-DataAgent-Automation-Lab**

New Notebook

Name *

Fabric-DataAgent-Automation

Location *

Fabric DA

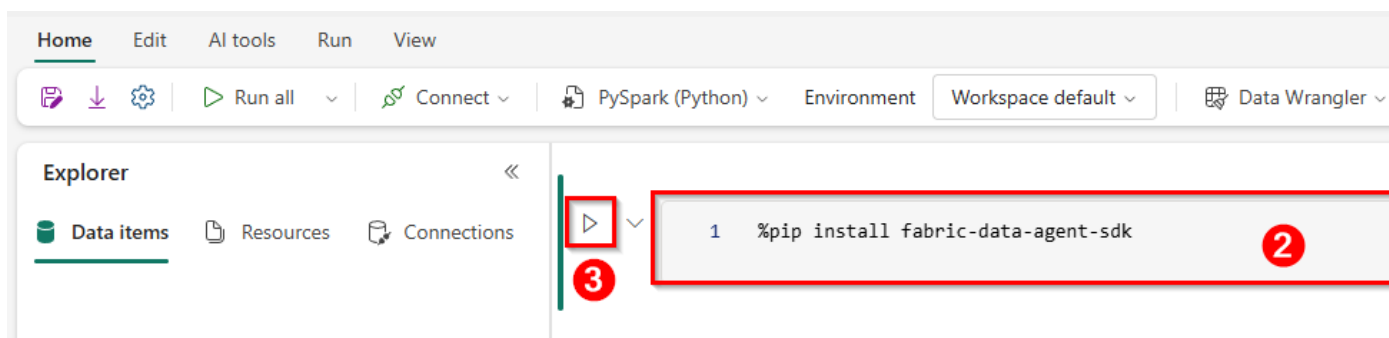
Create

Cancel

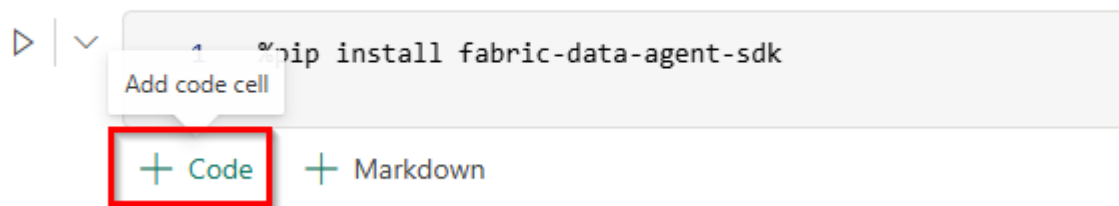
Task 2: Install SDK and Initialize Environment

1. Install the SDK, replace the content of a cell, add the following code and run the cell.

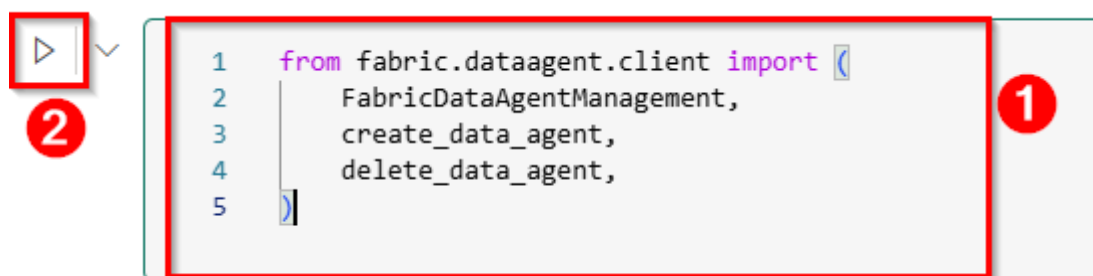
```
%pip install fabric-data-agent-sdk
```



2. Create a new cell, add the following code to Import required libraries.



```
from fabric.dataagent.client import (  
    FabricDataAgentManagement,  
    create_data_agent,  
    delete_data_agent,  
)
```



Task 3: Create and Configure Data Agent

Add a new cell for each step mentioned below copy and paste the code and run the cell to complete the specified step:

1. Create a new Data Agent

```
data_agent_name = "data_agent_adwork"  
data_agent = create_data_agent(data_agent_name)
```

2. Verify initial configuration

```
data_agent.get_configuration()
```

3. (Optional) Initialize existing Data Agent

```
data_agent = FabricDataAgentManagement(data_agent_name)
```

4. Add AI instructions

```
data_agent.update_configuration(  
    instructions="You are a helpful assistant, help users with their questions",  
)
```

5. Verify updated configuration

```
data_agent.get_configuration()
```

Task 4: Add and Publish Datasource

Add a new cell for each step mentioned below copy and paste the code and run the cell to complete the specified step:

1. Add Lakehouse datasource

```
lakehouse_name = "AdventureWorks"  
data_agent.add_datasource(lakehouse_name, type="lakehouse")
```

2. Verify datasource

```
data_agent.get_datasources()
```

3. Publish the Data Agent

```
data_agent.publish()
```

Task 5: Explore and Manage Datasource Schema

Add a new cell for each step mentioned below copy and paste the code and run the cell to complete the specified step:

1. Get datasource reference

```
datasource = data_agent.get_datasources()[0]
```

2. View available tables

```
datasource.pretty_print()
```

3. Select required tables

```
datasource.select("dbo", "dimcurrency")  
datasource.select("dbo", "dimemployee")
```

Note: As this feature is in Preview, this step might not work. Open your data agent created in previous step in fabric portal, for the newly added data source select the above-mentioned table manually.

4. Verify table selection

```
datasource.pretty_print()
```

5. Unselect a table

```
datasource.unselect("dbo", "dimcurrency")
```

6. Verify updated selection

```
datasource.pretty_print()
```

Task 6: Add Datasource Instructions

Add a new cell for each step mentioned below copy and paste the code and run the cell to complete the specified step:

1. Define instructions

```
ds_notes = "When asked about employees use the dimemployee table to retrieve detailed information"
```

2. Update datasource configuration

```
datasource.update_configuration(instructions=ds_notes)
```

3. Verify instructions

```
datasource.get_configuration()["additional_instructions"]
```

Task 7: Add and Manage Few-Shot Examples

1. Add few-shot example

```
example_dict = {  
    "How many employees are there in the company?":  
    "SELECT COUNT(*) AS NumberOfEmployees FROM dbo.dimemployee"  
}
```

```
datasource.add_fewshots(example_dict)
```

2. Verify few-shot examples

```
datasource.get_fewshots()
```

3. Remove few-shot example

```
datasource.remove_fewshot("REPLACE_WITH_YOUR_ID")
```

4. Verify removal

```
datasource.get_fewshots()
```

Task 8: Delete Data Agent

Add a new cell for each step mentioned below copy and paste the code and run the cell to complete the specified step:

1. Delete the Data Agent

```
delete_data_agent(data_agent_name)
```